

REVISION HISTORY			
REV	DESCRIPTION	DWN	CHK
1	ENGR RELEASE PER		

sUPI PAYLOAD BOARD

CONTENTS

TITLE	SHEET
BLOCK DIAGRAM	2
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IDEAL DIODE	5
RAIL DISCHARGE	6
DATA SWITCHES	7

RELATED PARTS			
TYPE	NUMBER	CFG	REV
PWB	100068610	001	1
PWA	100068622	001	1
EBOM	100068622	001	1

DRAWING MEETS REQUIREMENTS OF MIL-STD-31000		CONTRACT NO.	
DO NOT SCALE THIS DRAWING		DWN	
UNLESS OTHERWISE SPECIFIED CAPACITORS ARE IN MICROFARADS RESISTORS ARE IN OHMS DIMENSIONS ARE IN INCHES		CHK	
		APVD	
		APVD	
		APVD	
TOLERANCE ON DECIMALS ANGLES		APVD	
± .01 ± .1°		APVD	
± .005		BY	DIRECTION
		DATE	

SIZE	D	CAGE CODE	51993	DRAWING NO.	100046137	REV	1
SCALE	1:1	FILENAME: 100068622-001_Rev1.kicad_sch				SHEET: 1/7	

DRAPER®

555 TECHNOLOGY SQUARE
CAMBRIDGE, MA 02139-3563

DESIGN NOTES

- 1

VBATT IS EXPOSED TO PAYLOAD AFTER IDEAL DIODE TO PREVENT ADDITIONAL PAYLOAD CAPACITANCE FROM TRANSFERRING CHARGE BACK TO THE PLATFORM
- 2

CONNECTOR PINOUT IS MIRRORED FROM PLATFORM

- FID1

FIDuc1el
- FID2

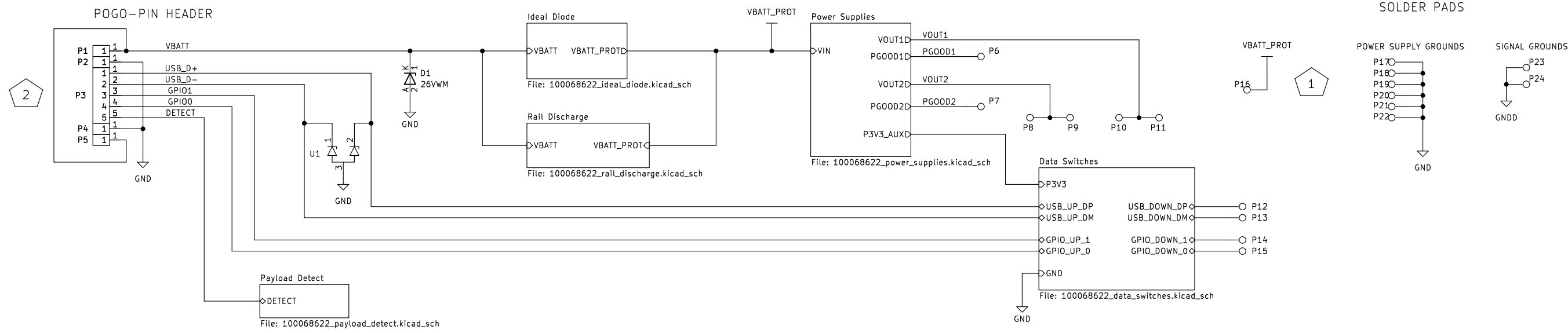
FIDuc2el
- FID3

FIDuc3el
- FID4

FIDuc4el
- FID5

FIDuc5el
- FID6

FIDuc6el



THIS DOCUMENT IS SUBJECT TO THE RESTRICTIONS ON SHEET 1 OF THIS DOCUMENT.				
SIZE	CAGE CODE	DRAWING NO.	REV	
D	51993	100046137	1	
SCALE	FILENAME: 100068622_block_diagram.kicad_sch			SHEET: 2/7
1:1	SHEET: Block Diagram			1

LAYOUT GUIDANCE

- 1

PLACE NEAR U1 PINS 8 AND 9
- 2

PLACE NEAR U1 PINS 6 AND 7
- 3

MARK SILKSCREEN AS "3V3"
- 4

MARK SILKSCREEN AS "5V"
- 5

MARK SILKSCREEN AS "7V4"
- 6

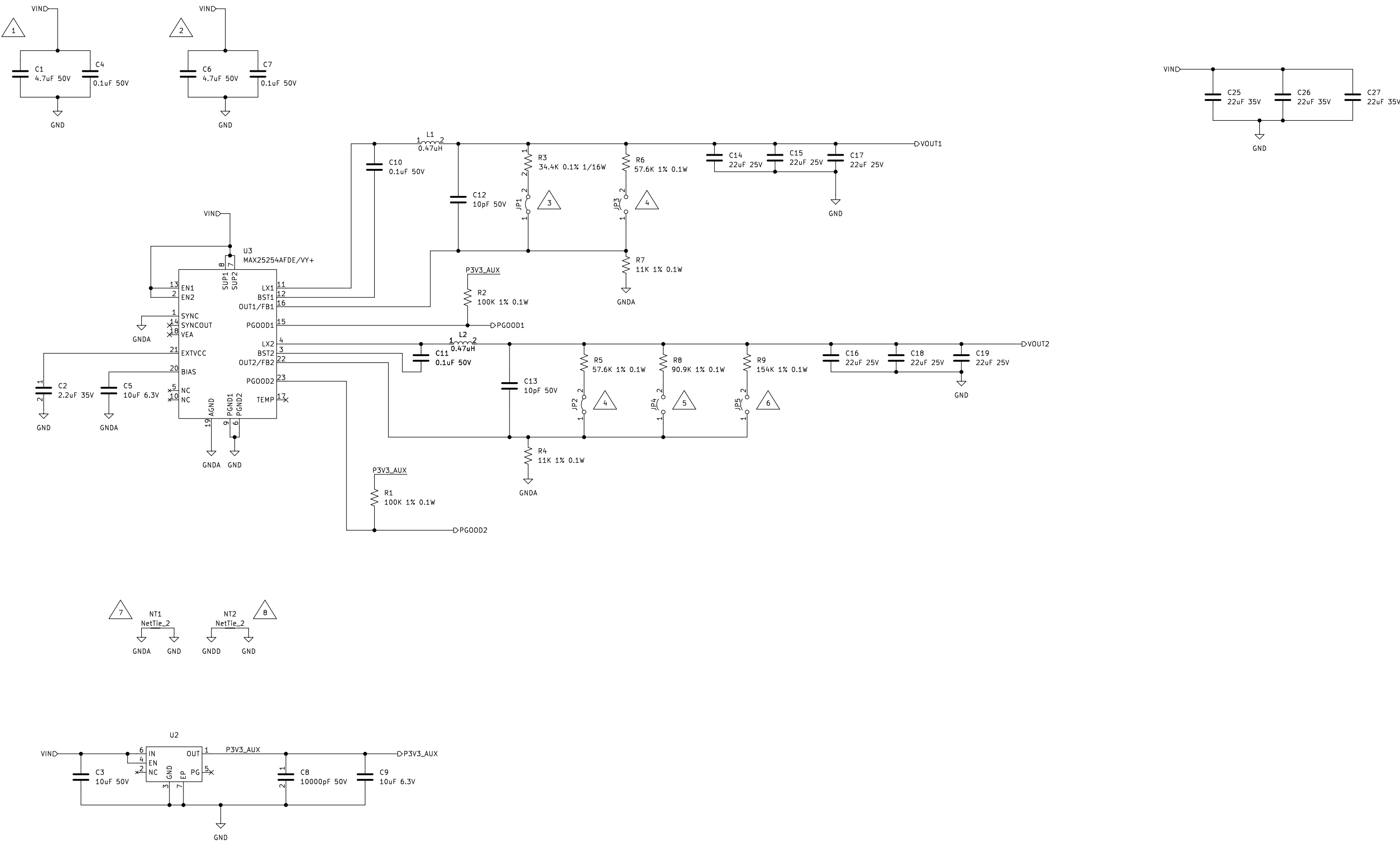
MARK SILKSCREEN AS "12V"
- 7

CONNECT GND TO GND4 AT A SINGLE POINT AWAY FROM NOISE
- 8

CONNECT GND TO GND0 AT A SINGLE POINT AWAY FROM NOISE

REVISION HISTORY				
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POWER SUPPLIES



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SIZE	CAGE CODE	DRAWING NO.	REV	
D	51993	100046137	1	
SCALE	FILENAME: 100068622_power_supplies.kicad_sch			
1:1	SHEET: Power Supplies			SHEET: 3/7

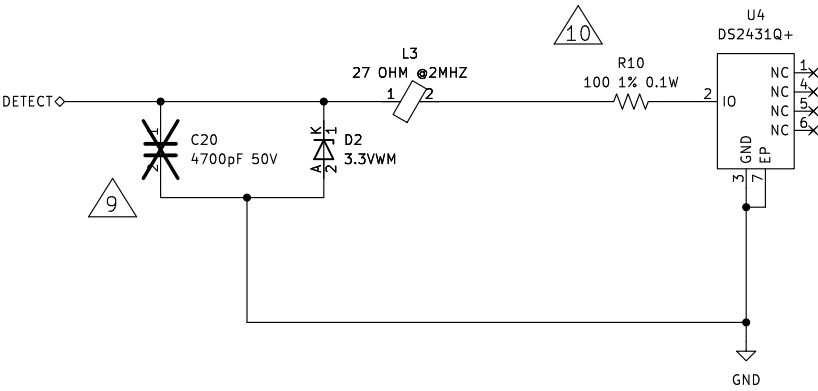
LAYOUT GUIDANCE

- 9

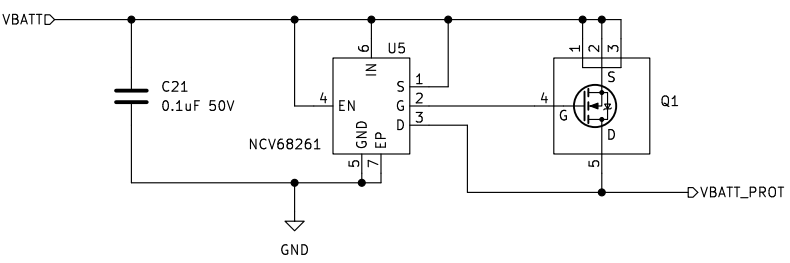
PLACE C26 AND D1 NEXT TO P12
- 10

PLACE R8 NEXT TO U2

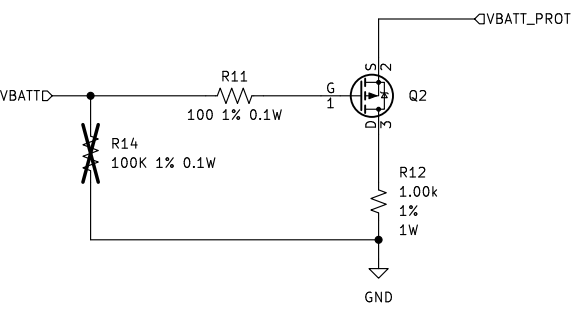
PAYLOAD DETECT



IDEAL DIODE



RAIL DISCHARGE



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SIZE	CAGE CODE	DRAWING NO.	REV
D	51993	100046137	1
SCALE	FILENAME: 100068622_rail_discharge.kicad_sch		
1:1	SHEET: Rail Discharge		SHEET: 6/7

REVISION HISTORY			
REV	DESCRIPTION	DWN	CHK APVD
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